SINE TESTING + TRANSIENT SDOF EXAMPLE / ABOUT 10 MINUTES

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The resonance your sweep never had time to build

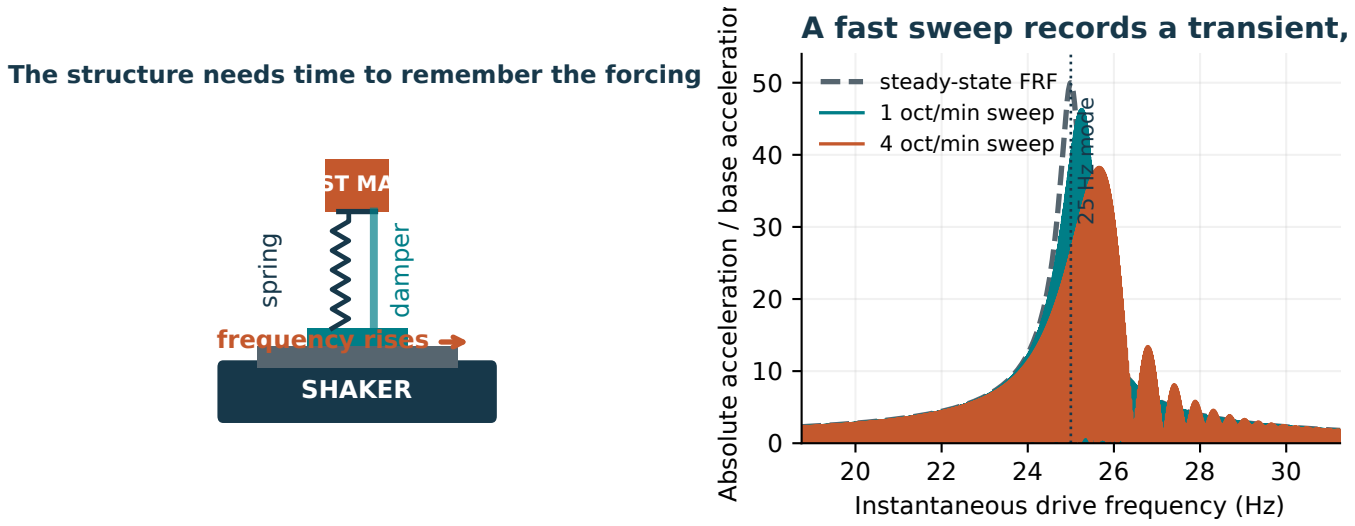


Figure 1. Original conceptual illustration and numerical response of a linear 25 Hz SDOF. The curves are illustrative calculations, not measured data or a test requirement.

A swept-sine plot looks like a frequency-response function, which makes it tempting to read every ordinate as steady state. But the shaker never holds most frequencies. Near a lightly damped resonance, energy needs time to accumulate; if frequency moves on too quickly, the response arrives late, peaks below the steady value, and reaches its maximum after the drive has crossed the natural frequency.

NASA engineers were documenting this problem in 1960. Reed, Hall and Barker showed that a frequency sweep produces a time-varying response: the measured resonant maximum is reduced and its frequency shifts in the sweep direction. Their point was measurement fidelity, but the same physics matters when a sine sweep is itself the verification environment. [1]

Today’s objective is to connect sweep rate, damping and modal frequency to ring-up time; reconstruct a transparent SDOF example; and separate a mission-shaped transient sweep from a resonance survey or dwell intended to characterize steady response.

01 / THE OSCILLATOR CARRIES MEMORY THROUGH RESONANCE

The instantaneous frequency is not the whole state

Let $y(t)$ be base motion and $z(t)=x(t)-y(t)$ the relative displacement of a linear SDOF. The base-driven equation is

$$\ddot{z}+2\zeta\omega_n\dot{z}+\omega_n^2 z=-\ddot{y}$$

At a fixed harmonic frequency ω, the steady-state absolute-acceleration transmissibility is

$$T_a(r)=\sqrt{\frac{1+(2\zeta r)^2}{(1-r^2)^2+(2\zeta r)^2}}\,,\qquad r=\frac{\omega}{\omega_n}$$

At resonance and low damping, $T_a(1) \approx 1/(2\zeta)=Q$. That familiar peak assumes the forcing has persisted long enough for the homogeneous transient to decay. The modal amplitude forgets its initial condition on the approximate envelope time constant

$$\tau=\frac{1}{\zeta\omega_n}=\frac{1}{2\pi\zeta f_n}$$

One time constant is only 63% of the way from the initial amplitude toward a new steady envelope; several are needed for close agreement. During a sweep, the target steady amplitude and phase change continuously, so the response always chases a moving solution.

For a logarithmic sweep rate S in octaves per minute,

$$\frac{\mathrm{d}f}{\mathrm{d}t}=\frac{S\ln 2}{60}f$$

A lightly damped half-power bandwidth is approximately $2\zeta f_n$. The time spent crossing that bandwidth is therefore

$$t_{\mathrm{HP}}\approx\frac{120\zeta}{S\ln 2}$$

This screening quantity is independent of modal frequency for a logarithmic sweep, while τ shrinks as frequency rises. Their ratio—not sweep rate alone—indicates whether a mode has time to build.

02 / WORKED EXAMPLE: THE SAME 1 g SWEEP, FOUR ANSWERS

A 25 Hz mode with 1% damping

Consider an illustrative linear SDOF with f_n=*25 Hz*, ζ=*0.01*, and a 1 g peak base-acceleration sine swept upward from 12.5 to 37.5 Hz. The steady resonant transmissibility is 50.01, so a sufficiently long dwell would approach 50.01 g peak. The response is calculated by direct time integration of the governing equation; no test data or proprietary model is used.

$$\tau = \frac{1}{2\pi(0.01)(25)} = 0.637 \text{ s}$$

At 4 octaves/minute, the sweep spends only

$$t_{HP} \approx \frac{120(0.01)}{4\ln 2} = 0.433 \text{ s} = 0.68\tau$$

inside the nominal half-power bandwidth. The oscillator cannot reach the steady resonant amplitude before the frequency moves on.

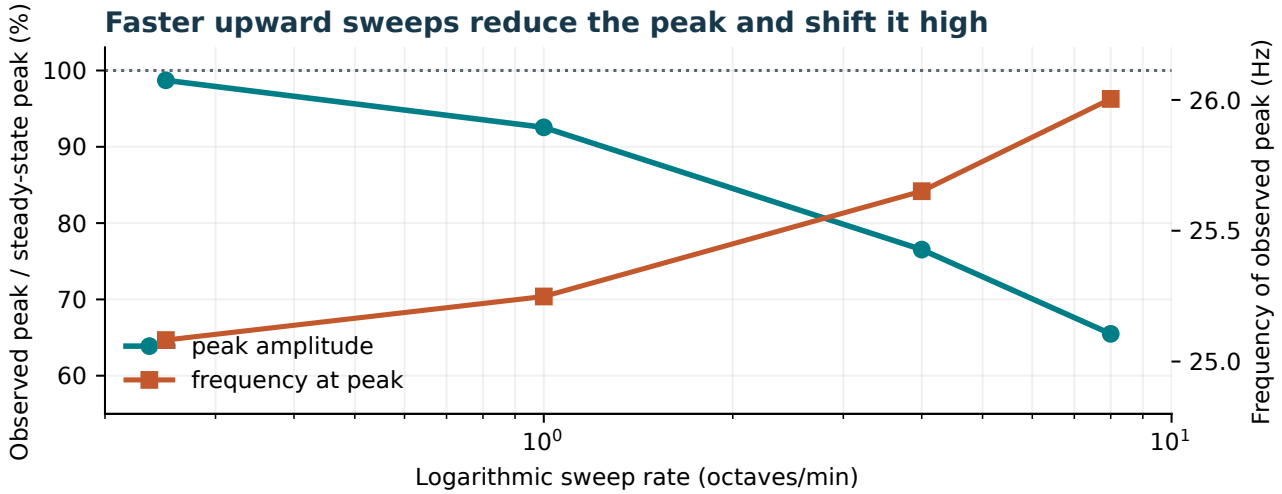


Figure 2. Original numerical integration. Peak amplitude is normalized by the 50.01 g steady-state result. Frequency shift is for an upward sweep.

Sweep	Peak response	Peak frequency	vs. steady
0.25 oct/min	49.4 g	25.08 Hz	98.7%
1 oct/min	46.3 g	25.25 Hz	92.5%
4 oct/min	38.3 g	25.65 Hz	76.5%
8 oct/min	32.7 g	26.00 Hz	65.5%

Table 1. Illustrative transient results. Slower is not automatically “more correct”; it is closer to steady state and applies more resonant cycles. Correctness depends on what the test is intended to represent.

03 / SWEEP RATE IS PART OF THE LOAD DEFINITION

Frequency history changes both peak and exposure

The ratio of bandwidth-crossing time to decay time is a useful pretest screen:

$$\frac{t_{\mathrm{HP}}}{\tau} \approx \frac{240\pi\zeta^2 f_n}{S\ln 2}$$

It reveals three practical trends. Lower-frequency modes have more difficulty ringing up during the same logarithmic sweep. Lighter damping hurts twice: it raises the steady resonance and narrows the band while lengthening memory. Faster sweeps further suppress and shift the response. This ratio is a heuristic, not an acceptance threshold; the full transient solution remains the comparison when peaks matter.

Direction matters too. In the same example, a 4 oct/min downward sweep peaks at 24.34 Hz rather than 25.65 Hz for the upward sweep, with nearly the same amplitude. For a perfectly linear SDOF, the peak shifts with the sweep direction. If up- and down-sweep amplitudes differ materially after accounting for control and initial conditions, suspect level-dependent damping or stiffness, friction, joint slip, preload change, contact, thermal drift, or another nonlinearity before averaging the curves together.

Sweep rate also sets cycle count. For a logarithmic sweep, the cycles accumulated between f_1 and f_2 are

$$N = \int_{t_1}^{t_2} f(t)\,\mathrm{d}t = \frac{60\,(f_2 - f_1)}{S\ln 2}$$

So reducing a sweep from 4 to 1 oct/min applies four times as many cycles through every frequency interval. That may improve modal characterization while also increasing fatigue exposure and dwell-like load at resonances. The choice is therefore not merely a data-resolution setting.

Lorenzo’s 1970 NASA variable-rate study made this trade explicit: sweep quickly where little dynamic information is present and slow near resonance. In a simulated lightly damped, double-resonance system, the method improved acquisition time or accuracy compared with a uniform fast sweep. The established lesson is adaptive allocation of time; the exact performance numbers belong to that study’s system and controller. [2]

04 / DESIGN THE TEST AROUND ITS PURPOSE

Survey, qualification and mission simulation are different jobs

For a resonance survey, choose a low level, slow enough rate, adequate sampling and suitable tracking/filter time constants to identify frequencies, shapes and damping without driving nonlinear behavior. If peak accuracy matters, check selected resonances with stepped sine or dwell and compare upward and downward sweeps. The response estimator, controller averaging and frequency resolution must settle faster than the structure is being traversed.

For mission simulation, preserve the duration and rate of change that create the relevant response. GSFC-STD-7000B specifies 4 oct/min for certain covered ELV payload sine tests to simulate flight sine-transient vibration, with lower rates where sustained pogo-like vibration must be represented. It also says subsystem/component sweep rates should match spacecraft-level rates and that base-drive predictions should be corrected if the analysis omitted sweep-rate effects. Those are GSFC program criteria within the standard's scope, not universal values. [3]

For fixture and low-level checkout, MSFC-STD-3676B gives 1 oct/min for the specific fixture/specimen resonance surveys it defines. That document is an MSFC technical standard and repeatedly directs users to verify the current version; its number is evidence of one scoped practice, not a rule for every facility or article. [4]

For analysis/test correlation, drive the FEM with the actual chirp history in direct or modal transient response, using the test's sweep direction, rate, amplitude schedule, notches and ramping. Compare synchronized time/frequency response, not only a steady-state SOL 111 envelope. Include fixture/interface dynamics, controller location, damping sensitivity and nonlinear joints where they can govern. A steady FRF remains valuable as the asymptote and for planning dwell frequencies; it is not automatically the expected sweep trace.

For verification closure, record actual instantaneous frequency, control motion, force and critical responses. Confirm notches and abort limits in low-level runs. NASA's spacecraft-level vibration review notes that sine and random strategies differ across centers but both use force and response limiting to protect against unrealistic test loads. [5]

05 / TAKEAWAYS, REFLECTION AND SOURCES

Three takeaways

1. A swept-sine trace is a transient response. Near resonance, finite sweep rate suppresses the peak and shifts it in the sweep direction relative to the steady FRF.

2. Ring-up depends on sweep rate, damping and frequency together. Compare the time inside modal bandwidth with $\tau=1/(\zeta\omega_n)$; use transient analysis when the result drives a decision.

3. The right rate follows the test objective. Surveys may slow near resonance; mission simulations preserve event timing; qualification plans must count added resonant cycles and respect scoped program criteria.

A short reflection

A steady-state SOL 111 predicts 50 g at a 25 Hz, 1%-damped mode, while a 4 oct/min test peaks near 38 g. Does the test prove the model is 24% high?

Answer: No. The illustrative transient calculation predicts about 38.3 g for that sweep before any model error. First run the actual chirp through the model and compare like with like. Then investigate damping, boundary conditions, fixture/control dynamics and nonlinearity. Use a dwell or slower local sweep if the objective is the steady resonant peak.

References

[1] W. H. Reed III, A. W. Hall and L. E. Barker Jr., Analog Techniques for Measuring the Frequency Response of Linear Physical Systems Excited by Frequency-Sweep Inputs, NASA TN D-508, October 1960.

[2] C. F. Lorenzo, Variable-Sweep-Rate Testing: A Technique to Improve the Quality and Acquisition of Frequency Response and Vibration Data, NASA TN D-7022, December 1970.

[3] GSFC-STD-7000B, General Environmental Verification Standard for GSFC Flight Programs and Projects, April 28, 2021. See Section 2.4.3; scoped GSFC criteria.

[4] MSFC-STD-3676B, Development of Vibroacoustic and Shock Design and Test Criteria, effective March 15, 2017. See Section 7.1.3; verify current applicability.

[5] S. Gordon et al., “Benefits of Spacecraft Level Vibration Testing,” Aerospace Testing Seminar, October 2015. Public NASA technical paper.

[6] ECSS-E-ST-10-03C Rev.1, Space Engineering - Testing, May 31, 2022. Includes resonance-search provisions; applies when invoked in an ECSS project.

Figures 1–2 and Table 1 are original illustrative calculations. Numerical integration uses a linear base-driven SDOF, constant 1% damping and a 1 g peak chirp; it is not measured evidence. This run could not verify or select GPT-6 Astra.